

KHOA (JAMES) HA

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EDUCATION

University of Manitoba

Computer Science Honours – Bachelor of Computer Science

May 2027

GPA: 4.25/4.5

- **Relevant coursework:** Data Structures and Algorithms, Software Development, Object-Oriented Programming

TECHNICAL SKILLS

Languages: JavaScript, C++, SQL, Typescript, HTML/CSS, R, Java, Python

Frameworks/Libraries: React, Express, NextJs, Tailwind CSS, NestJs, FastAPI, GraphQL

Databases: MySQL, MongoDB, Firebase, Redis

Cloud & DevOps: Docker, CI/CD, VPS Deployment, Github Action, AWS (S3, EC2)

Developing Tools/ Others: Git, Postman, RStudio

WORK EXPERIENCE

Computing Councils of Canada

Backend Developer Volunteer

Feb 2025 – Present

Winnipeg, Canada (remote)

- Developed and maintained a secure, scalable backend infrastructure with Fastify and NestJS, improving system reliability and site performance by 35%
- Integrated Google Calendar API for automated event scheduling and optimized real-time WebSocket communication for live help desk support serving 100+ students across Canada
- Improved API latency from 1200ms to 50ms using Redis caching

Geometric, Approximation and Distributed Algorithms Lab

Undergraduate Research Assistant

May 2025 – Aug 2025

Winnipeg, Canada (hybrid)

- Explored 5+ variations of the Cops and Robbers game on 10+ graphs, including strategic dynamics and rule modifications.
- Used greedy optimization to approximate minimal cop sets, improving pursuit efficiency by 25–30% on dense graphs.
- Analyzed lower/upper bounds in various types of graph, improving the time complexity for the Cops to capture the Robbers

NPH Digital

Software Developer Engineer Intern

May 2025 – Aug 2025

Long An, VietNam (remote)

- Developed an AI bot designed to support businesses by automating interactions and enhancing customer engagement, improving response efficiency by 40%
- Designed flexible tools for configuring conversational logic, supporting integration across 4+ platforms
- Implemented a full platform integration system, as measured by onboarding 50+ customers, by designing and deploying seamless API integrations with multiple third-party platforms.

PROJECTS

FrostByte – Hackathon Winner

Expo, Typescript, Firebase, Algorithm

Jan 2026

- Developed a feature enabling unhoused individuals to locate nearby warming shelters and navigate via shortest-path routing by introducing Dijkstra algorithm on a simulated map.
- Implemented REST APIs and deployed a Firebase serverless backend, enabling real-time alerts from unhoused individuals to volunteers.

SnazzyFits – Ecommerce App

NextJs, NestJs, OAuth, MySQL, Redis, VPS, Docker, Cloudinary, WebSocket, Github Action, GraphQL

Aug 2025 – Sep 2025

- Developed a distributed, containerized ecommerce platform with modular NestJS services and Redis caching for efficient product and transaction data handling.
- Integrated Google OAuth for secure authentication and built a suspicious login detection system using IP, device, and location analysis.
- Implemented Stripe and Sepay payment systems for transaction support, and real-time chat for customer support.
- Designed user behavior analytics to improve product recommendations and integrated Telegram/Discord bots with an LLM pipeline for AI-powered customer interaction.

PDFfinder – PDF chatbot

ReactJs, Tailwindcss, FastAPI, Langchain, AWS S3, Restful API, ChromaDB

Dec 2024 – Jan 2025

- Spearheaded the design and development of key components, focusing on user-centric interfaces and functionality
- Implemented RAG system allowing users to upload a PDF file and retrieves data in the file according to user's query
- Utilized AWS S3 for efficient PDF file storage and EC2 instances to host ChromaDB alongside a FastAPI server.

ACHIEVEMENTS & INVOLVEMENTS

IEEEExtreme 18.0 Programming Competition

Team Leader, Data Structure and Algorithms

Oct 2024

- Led a team of 3 and achieved 1st place at the University of Manitoba and 6th place across Canada